Uniform two-parameter selection in critical raw gradient descent

Abstract

For a three-variable polynomial objective, exact simultaneous gradient descent at a critical stepsize selects a minimizer according to a uniform square-root law. We prove all-time capture and point convergence for a two-parameter family, allowing the nonzero transverse perturbation to be arbitrarily small relative to the tangential displacement. The transverse coordinate never vanishes at finite time and alternates in sign. A stopped-orbit estimate controls the entire accumulated error, including the potentially long residence near an unstable minimizer. The limiting-point map is discontinuous along the positive tangential axis. The argument is analytic and requires no numerical prefix or uniform convergence-time estimate.

Keywords: critical gradient descent; minimizer selection; square-root selection law; uniform capture; limiting-point discontinuity

2020 Mathematics Subject Classification. Primary 37N40; Secondary 65K10, 39A30.

1 The exact map and its context

Consider gradient descent with stepsize $1/30$ on the nonnegative objective

$$f(a,b,c) = 50a^2b^2 + \frac{(ac-1)^2}{200}. \tag{1}$$

Set

$$u = \frac{10}{3}, \qquad h = \frac{1}{3000}, \qquad A = \sqrt{\frac{3}{5}}, \qquad C = A^{-1}, \qquad \lambda = \frac{9}{68}. \tag{2}$$

The exact simultaneous update is

$$\begin{aligned} a_{n+1} &= a_n - ua_nb_n^2 - hc_n(a_nc_n - 1), \\ b_{n+1} &= (1 - ua_n^2)b_n, \\ c_{n+1} &= c_n - ha_n(a_nc_n - 1). \end{aligned} \tag{3}$$

Every right-hand side uses the old state. There is no clipping, projection, cutoff, rounding or rescaling. Differentiating (1) verifies (3) directly. Its positive minimizer branch is $(a, 0, a^{-1})$, $a > 0$. At $(A, 0, C)$ the transverse multiplier is $1 - uA^2 = -1$.

The motivating optimization setting is the final NeurIPS 2025 paper of Liu, Zhang, Du and Zhao [1]. Its Section 3 theoretical algorithm is projected, Appendix E clips the noise coordinate, and Appendix G treats unconstrained experiments. Here we study the raw polynomial map (3) itself. The initialization family below is specified for this analysis; it is not attributed to that paper or identified with a figure seed. No claim of originality, global clipping removal, or coverage of the source's full initialization set is made.

2 Uniform selection theorem

For real parameters s, b near zero, initialize

$$(a_0, b_0, c_0) = (A + s, b, (A + s)^{-1}). \tag{4}$$

The letter b without an index denotes the initial parameter.

Theorem 1. *There are constants $\delta > 0$, $B < \infty$, and $R < \infty$ such that, if $|s| + |b| < \delta$ and $b \neq 0$, the orbit (3)–(4) has the following properties.*

1. *For every $n \geq 0$, $a_n, c_n > 0$ and*

$$|a_n - A| + |b_n| + |c_n - C| \leq B\sqrt{s^2 + b^2}.$$

2. *At every finite time, $b_n \neq 0$ and*

$$\operatorname{sign}(b_n) = (-1)^n \operatorname{sign}(b).$$

3. *For a strictly subcritical positive value $0 < a_\infty < A$,*

$$(a_n, b_n, c_n) \longrightarrow (a_\infty, 0, a_\infty^{-1}),$$

   which is a global minimizer of (1).

4. *With $\rho = \sqrt{s^2 + \lambda b^2}$,*

$$|a_\infty - A + \rho| \leq R(s^2 + b^2), \tag{5}$$
$$|a_\infty^{-1} - C - A^{-2}\rho| \leq R(s^2 + b^2). \tag{6}$$

   These bounds are uniform with no lower bound on $|b|/|s|$.

5. *For each fixed admissible initial point, there are $J < \infty$, $\theta \in (0, 1)$ and an integer $N \geq 0$ such that, for $n \geq N$,*

$$|a_n - a_\infty| + |b_n| + |c_n - a_\infty^{-1}| \leq J\theta^{n-N}.$$

The constants J, θ, N may depend on (s, b). No uniform rate or convergence time as $b \to 0$, and no amplitude monotonicity from time zero, is asserted.

3 Coordinates and exact identities

Put $v = b_n^2$ for the current squared transverse coordinate, and suppress time indices for one-step formulas. Write

$$D = a^2 - c^2, \quad e = ac - 1, \quad G(q) = q^2 - q^{-2}, \quad q = G^{-1}(D), \quad x = q - A. \tag{7}$$

The function G is strictly increasing from $(0, \infty)$ onto $\mathbb{R}$ since $G'(q) = 2q + 2q^{-3} > 0$. Near (A, C), these are smooth coordinates for the positive (a, c) component. In fact, from $D = G(A + x)$,

$$a^2 = \frac{D + \sqrt{D^2 + 4(1+e)^2}}{2}, \qquad a > 0, \qquad c = \frac{1+e}{a}. \tag{8}$$

Consequently $a = q + O(e)$ and $c = q^{-1} + O(e)$, uniformly in a fixed small neighborhood. The initialization gives exactly

$$x_0 = s, \qquad e_0 = 0, \qquad v_0 = b^2. \tag{9}$$

A superscript + denotes a simultaneous step of (3). Direct expansion yields

$$D^+ - D = -2ua^2v + u^2a^2v^2 + 2uhacev - h^2De^2, \tag{10}$$

$$e^+ = Qe - u(1+e)v, \tag{11}$$

$$v^+ = Mv, \qquad M = (1 - ua^2)^2, \tag{12}$$

$$Q = 1 - h(a^2 + c^2) + uha^2v + h^2(1+e)e. \tag{13}$$

In particular, the linear residual term cancels in $D^+ - D$. These identities refer only to the raw map, with no modified dynamics.

4 Uniform local estimates

4.1 Control of the residual

At the reference point $(x, e, v) = (0, 0, 0)$,

$$M = 1, \qquad Q = 1 - \kappa, \qquad \kappa = h(A^2 + C^2) = \frac{17}{22500}.$$

Choose a fixed sufficiently small coordinate neighborhood

$$\mathcal{U} = \{|x| < r,\ |e| < r,\ 0 \le v < r^2\}.$$

By continuity, throughout it a, c are positive, $1 - ua^2$ is negative and bounded away from zero, and, for a finite constant U,

$$|Q| \le 1 - \frac{3\kappa}{4}, \qquad M \ge 1 - \frac{\kappa}{4}, \qquad u|1+e| \le U. \tag{14}$$

Fix $K \geq 2U/\kappa$. If a current state in this neighborhood satisfies $|e| \leq Kv$, then (11)–(14) imply

$$|e^+| \leq \left[\left(1 - \frac{3\kappa}{4}\right) K + U\right] v \leq K\left(1 - \frac{\kappa}{4}\right) v \leq Kv^+. \tag{15}$$

This is a one-step propagation estimate. It does not assume that the trajectory stays local; that fact is proved below. It uses no division by b and includes $v = 0$.

4.2 Slow drift and energy error

All constants from now on are uniform on a possibly smaller fixed neighborhood, restricted to $|e| \leq Kv$. The symbol $O(\cdot)$ means an absolute bound by a constant times its nonnegative argument on this set, independent of the initial parameters and of time.

Since $e = O(v)$ and $a = q + O(v)$, (10) becomes

$$D^+ - D = -2uq^2v + O(v^2).$$

Taylor's theorem applied to G^{-1} therefore gives

$$x^+ = x - k(x)v + O(v^2), \qquad k(x) = \frac{2uq^2}{G'(q)} = \frac{uq^5}{q^4 + 1}, \quad q = A + x. \tag{16}$$

To justify this expansion before capture, shrink the neighborhood so that the raw image has positive a^+, c^+ and its D^+ lies in a fixed compact interval where the derivatives of G^{-1} are bounded. Continuity of the raw map, which fixes the reference point, supplies this condition. The terms involving ev and e^2 in (10) are both $O(v^2)$, so no stable-coordinate effect is omitted.

The function k is positive and smooth, with

$$k_0 = k(0) = \frac{15}{17}A, \qquad \frac{k_0}{2uA} = \frac{9}{68} = \lambda. \tag{17}$$

Also, expanding (12) with $a = A + x + O(v)$ and $uA^2 = 2$ gives

$$v^+ = v + 4uAxv + O((x^2 + v)v). \tag{18}$$

Define the positive quadratic expression

$$H = x^2 + \lambda v. \tag{19}$$

The leading terms in $H^+ - H$ are $-2k_0xv$ from the x coordinate and $4\lambda uAxv$ from the v coordinate. They cancel by (17). The remaining terms are bounded by a constant times $x^2v + v^2$, hence by a constant times Hv. Thus, for some $L < \infty$,

$$|H^+ - H| \leq LHv. \tag{20}$$

This is an error bound, not an exact conservation or monotonicity assertion. Positivity of k in (16), after one final neighborhood reduction, also gives fixed $d, D_1 > 0$ such that

$$dv \leq x - x^+ \leq D_1 v. \tag{21}$$

No monotonicity assumption on v is used.

5 A closed stopped-orbit argument

Put

$$t = \sqrt{H_0} = \sqrt{s^2 + \lambda b^2}.$$

We establish local capture without assuming the conclusion of the drift estimates. First fix the neighborhood and all constants K, L, d, D_1. Then choose the allowable t sufficiently small that every state with $H \le 4t^2$ and $|e| \le Kv$ lies well inside that neighborhood. This is possible because

$$|x| \le 2t, \qquad v \le 4t^2/\lambda, \qquad |e| \le 4Kt^2/\lambda.$$

The raw images of all these states also lie inside the neighborhood when t is small: the sets shrink uniformly to the fixed reference point and the map is continuous there. In particular, the image coordinates and (15) are legitimate. Require in addition

$$\frac{4Lt^2}{\lambda} \le \frac{1}{4}, \qquad \exp\left(\frac{L(1+\sqrt{5})t}{d}\right) \le 2. \tag{22}$$

Consider any finite prefix for which $H_i \le 4t^2$ and $|e_i| \le Kv_i$ for $0 \le i < n$. Its final update is legitimate by the preceding neighborhood choice. Equation (20) and (22) first imply $H_n \le 5t^2$. Telescoping the lower drift estimate, including the final step of this prefix, gives

$$\sum_{i=0}^{n-1} v_i \le \frac{x_0 - x_n}{d} \le \frac{(1+\sqrt{5})t}{d}, \tag{23}$$

since $x_0 = s \le t$ and $|x_n| \le \sqrt{5}t$. Applying the upper bound in (20) at these already justified previous states now yields

$$H_n \le t^2 \prod_{i=0}^{n-1}(1 + Lv_i) \le t^2 \exp\left(L \sum_{i=0}^{n-1} v_i\right) \le 2t^2. \tag{24}$$

The tube propagates by (15). Starting from (9), induction therefore improves the provisional $4t^2$ bound to $2t^2$ at every finite time. This proves actual all-time local capture and gives

$$H_n \le 2t^2, \qquad |e_n| \le Kv_n \qquad (n \ge 0). \tag{25}$$

With capture now established, (21) telescopes over arbitrary prefixes to give

$$\sum_{n=0}^{\infty} v_n \le \frac{(1+\sqrt{2})t}{d}. \tag{26}$$

Combining this with (20)–(25), we obtain a constant $L_1 < \infty$, independent of (s, b), such that

$$|H_n - t^2| \le \sum_{i<n} LH_i v_i \le L_1 t^3 \qquad (n \ge 0). \tag{27}$$

For example, $L_1 = 2L(1+\sqrt{2})/d$ is admissible. Reduce the allowable t further so $L_1 t \le 1/2$. Then $H_n \ge t^2/2$ for $t > 0$.

Finally, (25) gives $|x_n| \le \sqrt{2}t$, $|b_n| \le \sqrt{2/\lambda}\, t$, and $|e_n| \le 2Kt^2/\lambda$. The smooth inverse (8) gives the stated capture bound, since $t \le \sqrt{s^2+b^2}$. A single sufficiently small δ in the theorem enforces all preceding smallness conditions. In particular, none depends on a positive lower bound for $|b|/|s|$.

6 Convergence and selection

For $b \neq 0$, the local multiplier $1 - ua_n^2$ remains negative and bounded away from zero. Induction proves nonvanishing and sign alternation at every finite time. In particular, every $v_n > 0$.

By (21), x_n is decreasing, and by (25) it is bounded below. Let its limit be x_∞. Summability (26) implies $v_n \to 0$; the residual tube implies $e_n \to 0$. The coordinate inverse therefore proves

$$(a_n, b_n, c_n) \longrightarrow (A + x_\infty, 0, (A + x_\infty)^{-1}).$$

The first coordinate is positive by the small neighborhood choice, and the objective at the limit is zero.

Passing to the limit in (27) gives

$$|x_\infty^2 - t^2| \le L_1 t^3, \qquad x_\infty^2 \ge t^2/2 > 0. \tag{28}$$

Thus $x_\infty \neq 0$. If $x_\infty > 0$, then $a_\infty = A + x_\infty > A$ and $(1 - ua_\infty^2)^2 > 1$. For some $\eta > 0$, convergence would imply $v_{n+1} \ge (1+\eta)v_n$ at every sufficiently large n. Because $v_n > 0$, this contradicts $v_n \to 0$. Hence $x_\infty < 0$, proving strict side selection even for $s > 0$ and arbitrarily small nonzero b.

Using this sign and (28),

$$|x_\infty + t| = \frac{|x_\infty^2 - t^2|}{|x_\infty| + t} \le L_1 t^2.$$

As $t = \rho$ and $t^2 \le s^2 + b^2$, this proves (5). Taylor expansion of the reciprocal on a fixed positive interval gives

$$a_\infty^{-1} = C - A^{-2} x_\infty + O(x_\infty^2) = C + A^{-2}\rho + O(t^2),$$

with a uniform constant. Enlarging R proves (6).

For completeness, fix one admissible initial point. Since $0 < a_\infty < A$, the limiting absolute multiplier is strictly below one. Choose $\beta \in (0,1)$ and N such that $|1 - ua_n^2| \le \beta$ for all $n \ge N$. Then

$$\begin{aligned} |b_n| &\le |b_N| \beta^{n-N}, \\ v_n &\le v_N \beta^{2(n-N)}, \qquad |e_n| \le K v_N \beta^{2(n-N)}. \end{aligned}$$

The upper drift estimate gives

$$0 \le x_n - x_\infty \le D_1 \sum_{j \ge n} v_j \le \frac{D_1 v_N}{1-\beta^2}\, \beta^{2(n-N)}.$$

The inverse coordinate map is locally Lipschitz on the fixed neighborhood. The errors in a, c are therefore bounded by a constant times $|x_n - x_\infty| + |e_n|$. Together these inequalities establish the geometric tail with $\theta = \beta$ and finite J, completing the proof of Theorem 1.

7 Discontinuity and scope

Define the limiting-point map on the entire sufficiently small initial family, including $b = 0$. For $b = 0$, (4) has $e = 0$ and is exactly fixed, so its limit is $(A + s, 0, (A + s)^{-1})$.

Corollary 7.1. *For every sufficiently small fixed $s > 0$, this limiting-point map is discontinuous at $(s, 0)$. More precisely, after enlarging R if needed,*

$$\begin{aligned} A - s - Rs^2 &\le \liminf_{\substack{b \to 0 \\ b \neq 0}} a_\infty(s, b) \\ &\le \limsup_{\substack{b \to 0 \\ b \neq 0}} a_\infty(s, b) \le A - s + Rs^2 < A. \end{aligned}$$

Proof. For fixed $s > 0$, the quantity $\rho(s, b)$ tends to s as $b \to 0$. Apply the uniform inequality (5) and take lower and upper limits. Taking s sufficiently small that $Rs < 1$ makes the upper endpoint strictly less than A, whereas the value at $b = 0$ has first coordinate $A + s$. ∎

This argument does not assert that the punctured limit as $b \to 0$ exists for any fixed positive s; the lower and upper limiting bounds suffice. The potentially long time spent near the positive-s unstable minimizer is absorbed by the stopped-orbit sum (23), rather than bounded separately.

At $s = 0$, the theorem specializes to

$$a_\infty = A - \frac{3}{2\sqrt{17}}|b| + O(b^2), \qquad c_\infty = C + \frac{5}{2\sqrt{17}}|b| + O(b^2).$$

The two-parameter conclusion is stronger because it also covers $s > 0$ without any lower bound on the transverse perturbation. Its proof is self-contained and does not use a one-parameter convergence theorem as a premise.

All thresholds here are existential. The initialization has exactly $a_0 c_0 = 1$, and the map and stepsize are exactly as stated. The result does not automatically extend to arbitrary nearby three-dimensional initializations, perturbed stepsizes, finite-precision simulation, or clipped dynamics.

A Portable algebra check

The accompanying script `check_identities.py` expands the raw polynomial update with exact SymPy expressions. It checks (10), (11), and (12), the gradient-descent coefficients, $\kappa = 17/22500$, and $k_0/(2uA) = 9/68$. Run it with Python and SymPy installed:

```
python3 check_identities.py
```

Its successful output is `PASS: exact raw-map algebra identities`. This is an algebra check only. It supplies neither the all-time capture proof nor a numerical convergence certificate, and does not test an assumed Lyapunov identity. The analytic proof above supplies the uniform estimates and the infinite-time conclusions.

References

[1] Liu, Zhang, Du and Zhao. *A Minimalist Example of Edge-of-Stability and Progressive Sharpening*. Advances in Neural Information Processing Systems, 2025, final proceedings version. https://papers.nips.cc/paper_files/paper/2025/file/15eac388458e8a577f96edf7a40d0cbc-Paper-Conference.pdf.